

Research Notes on Batch normalization

Batch normalization

$$\hat{X}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

$$y_i = \gamma \hat{x}_i + \beta$$

>> Section 1

$\min_{w \sim \in \mathbb{R}^d} \text{OLS}(w) = \min_{w \sim \in \mathbb{R}^d} (E_{x,y}[(y - x^T w)^2]) = \min_{w \sim \in \mathbb{R}^d} (2u^T w + w^T S w)$
$$\{\tilde{w}\} = \min_{\{\tilde{w}\} \in \mathbb{R}^d} (E_{x,y}[(y - x^T \tilde{w})^2]) = \min_{\{\tilde{w}\} \in \mathbb{R}^d} (2u^T \tilde{w} + \tilde{w}^T S \tilde{w})$$
, where $u = E[-yx]$
$$S = E[xx^T]$$
. Since $w \sim \gamma w + \beta$
 $\{\tilde{w}\} = \gamma \{\tilde{w}\} + \beta$, the objective thus becomes

>> Section 2

$\|w_t\|_S^2 \leq f_{LH}(w_t, g_t) \leq S^{-1} 2 \leq (1 - \mu_L) 2 t \Phi^2 \gamma t^2 (\rho(w_0) - \rho^*)$
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 $f_{LH}(w_t, g_t) \leq S^{-1} 2 \leq (1 - \mu_L) 2 t \Phi^2 \gamma t^2 (\rho(w_0) - \rho^*)$. Combining these two inequalities, a bound could thus be obtained for the gradient with respect to $w \sim d$
 $\{\tilde{w}\}_T$:

>> Section 3

$w \sim d = \gamma T d w T d \parallel w T d \parallel S$
 $\{\tilde{w}\}_T = \gamma T d w T d \parallel w T d \parallel S$
 $\{\tilde{w}\}_T = \gamma T d w T d \parallel w T d \parallel S$. The GDNP algorithm thus slightly modifies the batch normalization step for the ease of mathematical analysis. It can be shown that in GDNP, the partial derivative of f_{LH} against the length component converges to zero at a linear rate, such that

>> Section 4

$y_i(k) = \gamma(k) x_i(k) + \beta(k)$
 $y_i(k) = \gamma(k) x_i(k) + \beta(k)$, where the parameters $\gamma(k)$ and $\beta(k)$ are subsequently learned in the optimization process. Formally, the operation that implements batch normalization is a transform $BN_{\gamma(k), \beta(k)}: x_{1...m}(k) \rightarrow y_{1...m}(k)$
 $BN_{\gamma(k), \beta(k)}: x_{1...m}(k) \rightarrow y_{1...m}(k)$ called the Batch Normalizing transform. The output of the BN

transform $y(k) = BN_{\gamma(k), \beta(k)}(x(k))$ is then passed to other network layers, while the normalized output $\hat{x}_i(k)$ remains internal to the current layer.

>> Section 5

Least-square problem With the reparametrization interpretation, it could then be proved that applying batch normalization to the ordinary least squares problem achieves a linear convergence rate in gradient descent, which is faster than the regular gradient descent with only sub-linear convergence. Denote the objective of minimizing an ordinary least squares problem as

>> Section 6

$\|w \sim f(w \sim d)\|_2 \leq (1 - \mu_L) 2 t d \Phi^2 (\rho(w_0) - \rho^*) + 2 - T s \zeta |b_t(0) - a_t(0)| \mu^2$
 $\|w \sim f(w \sim d)\|_2 \leq (1 - \mu_L) 2 t d \Phi^2 (\rho(w_0) - \rho^*) + 2 - T s \zeta |b_t(0) - a_t(0)| \mu^2$
 $\|w \sim f(w \sim d)\|_2 \leq (1 - \mu_L) 2 t d \Phi^2 (\rho(w_0) - \rho^*) + 2 - T s \zeta |b_t(0) - a_t(0)| \mu^2$, such that the algorithm is guaranteed to converge linearly. Although the proof stands on the assumption of Gaussian input, it is also shown in experiments that GDNP could accelerate optimization without this constraint.

>> Section 7

$\rho \sim (w) = w^T B w w^T A w$
 $\rho \sim (w) = w^T B w w^T A w$, where $B \in \mathbb{R}^d \times d$ is a symmetric matrix and $A \in \mathbb{R}^d \times d$ is a symmetric positive definite matrix. It is proven that the gradient descent convergence rate of the generalized Rayleigh quotient is

>> Section 8

$w \sim f_{LH}(w \sim) = c_1(w \sim) u + c_2(w \sim) S w \sim$
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 $w \sim f_{LH}(w \sim) = c_1(w \sim) u + c_2(w \sim) S w \sim$, where $c_1(w \sim) = E z[\phi(1)(z T w \sim)] - E z[\phi(2)(z T w \sim)](u^T w \sim)$
 $c_1(w \sim) = E z[\phi(1)(z T w \sim)] - E z[\phi(2)(z T w \sim)](u^T w \sim)$
 $c_2(w \sim) = E z[\phi(2)(z T w \sim)](u^T w \sim)$, and $\phi(i)$ is the i -th derivative of ϕ . By setting the gradient to 0, it thus follows that the bounded critical points $w \sim^*$ can be expressed as $w \sim^* = g^* S^{-1} u$, where g^* depends on $w \sim^*$ and ϕ .

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Combining this global property with length-direction decoupling, it could thus be proved that this optimization problem converges linearly. First, a variation of gradient descent with batch normalization, Gradient Descent in Normalized Parameterization (GDNP), is designed for the objective function $\min_{w \in \mathbb{R}^d \setminus \{0\}, \gamma \in \mathbb{R}} f_{LH}(w, \gamma)$ $\{\displaystyle \min_{w \in \mathbb{R}^d \setminus \{0\}, \gamma \in \mathbb{R}} f_{LH}(w, \gamma)\}$, such that the direction and length of the weights are updated separately. Denote the stopping criterion of GDNP as

>> Section 9

$(\partial_{\gamma} f_{LH}(w_t, a_t(T_s)))^2 \leq 2^{-T_s \zeta} |b_t(0) - a_t(0)|$ $\mu^2 \{\displaystyle (\partial_{\gamma} f_{LH}(w_t, a_t(T_s)))^2 \leq \frac{2^{-T_s \zeta}}{|b_t(0) - a_t(0)| \mu^2}\}$, where $a_t(0)$ $\{\displaystyle a_t(0)\}$ and $b_t(0)$ $\{\displaystyle b_t(0)\}$ are the two starting points of the bisection algorithm on the left and on the right, correspondingly. Further, for each iteration, the norm of the gradient of f_{LH} $\{\displaystyle f_{LH}\}$ with respect to w $\{\displaystyle w\}$ converges linearly, such that